EU AI Act Compliance Review

AI Act Compliance: Review of AURA Subsystem 02

Station: AI Security Officer (Regulatory Compliance & Security)
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Scope: FST Dialogue Manager, Noisy Channel Simulator | Legal basis: Regulation (EU) 2024/1689 (AI Act) | Review date: 2026-10-25

1. Executive Summary

Question	Answer
Is the FST a prohibited practice (Art. 5)?	No. The system does not use subliminal techniques or exploit vulnerabilities.
Is it a high-risk AI system (Art. 6, Annex III)?	No. Navigational simulation in a course setting is not listed in Annex III.
Do transparency duties apply (Art. 50)?	Yes, if the simulator is deployed to users, as it generates text interacting with humans.
Does the forced noise token violate safety rules?	No, provided the noise is explicitly declared as a channel constraint to the user.

Verdict: The FST Noisy Channel Simulator is compliant as an educational prototype. However, deploying a system that intentionally injects semantic noise (e.g., "Banana", "Labyrinth") into critical navigational readouts poses a severe operational hazard outside of the AI Act's purview.

2. Risk Classification Analysis

Prohibited practices (Art. 5) are not triggered. While the Noisy Channel injects random words, this is a simulated communications failure, not a "covert persuasion" or manipulative technique meant to alter human behavior.

High-risk (Annex III) is not triggered. Subsystem 02 does not manage critical national infrastructure, biometric categorization, or law enforcement data. It is a localized deterministic state machine.

Limited risk (Art. 50) applies. The system communicates directly with natural persons. Even though the FST is deterministic and not a neural network, the AI Act’s transparency obligations apply broadly to systems mimicking human dialogue.

3. Article 50 Transparency Checklist

Requirement	Status	Evidence / Remediation
Users must be told they are interacting with AI (Art. 50(1))	Gap	The INIT state must append: "You are talking to an automated FST system."
AI-generated text marked in a machine-readable way	N/A	The FST outputs hardcoded strings, not dynamically generated text.
Deep-fake disclosure (Art. 50(4))	Pass	No visual or audio manipulation is present.
Emotion recognition (Art. 50(3))	Pass	Not used. Regex only parses slots.

4. Security and Hazard Checks

The transition from probabilistic LLMs to deterministic FSTs eliminates Prompt Injection vulnerabilities (as verified in Stress Test 5). However, the Noisy Channel constraint introduces a unique hazard: Semantic Drift.

If an operator relies on the text output, the mandatory noise token (e.g., "Cosmological") might be misinterpreted as a valid parameter rather than channel distortion. A remediation strategy must be implemented to clearly delimit noise tokens from operational data (e.g., encapsulating noise in brackets: `[NOISE: Quantum]`).

5. Actions Before Final Deployment

- 1. Append the AI disclosure notice to the GREETING state (Art. 50(1)).
- 2. Add visual delimiters to the noise token injection function to prevent semantic drift.
- 3. Implement a reverse-edge transition in the FST state diagram to allow for slot backtracking without flushing the entire memory buffer.